

Replication Report:

Unconventional Superconductivity and Density Waves in Twisted Bilayer Graphene (Isobe, Yuan, Fu 2018)

TEXTURES-100 automated replication (loop-current class)

2026-07-19

1 Paper and scope

Target: H. Isobe, N. F. Q. Yuan, L. Fu, *Unconventional Superconductivity and Density Waves in Twisted Bilayer Graphene*, Phys. Rev. X **8**, 041041 (2018); arXiv:1805.06449v2. The paper introduces a **hot-spot patch renormalization-group (RG)** model for twisted bilayer graphene near the $n = 2$ Van Hove filling: nine momentum-conserving four-fermion couplings g_{ij} flow under one-loop RG (Eqs. 9–15), and RPA-resummed susceptibilities (Eqs. 16, 24) diagnose ordering instabilities with channel interaction strengths V_η (Eqs. 17–23).

Shared-kernel provenance / scope decision. The assigned TEXTURES-100 loop-current class ships a kagome tight-binding + Peierls-flux kernel (`loop_current_kagome_kernel.py`, built for Fernandes *et al.* arXiv:2502.16657). It was read first, as required. **It is out of scope for this paper’s core:** Isobe *et al.* has no tight-binding lattice, no Peierls flux, no Berry curvature/Chern number. The kernel’s `KagomeModel`, `chern_number`, `bond_current_and_charge`, and `triangle_flux_from_config` were therefore *not* imported. Only the conceptual “couplings $\rightarrow$ ordering-channel selection” idea (the kernel’s Box-2 `patch_leading_channel`) is thematically shared; the actual in-scope core (9-coupling RG) was reimplemented from the paper’s equations in `isobe2018_rg.py`. This is an honest class-mismatch flag, not a defect.

2 Machine-checkable claims and results

All numbers below are computed at runtime by `code/run_checks.py` and stored in `work/results.json`; none are transcribed from the paper’s figures.

1. **Eq. (9): frozen intravalley couplings.** g_{14}, g_{24}, g_{44} have identically zero RG beta-function. Verified $\max |\dot{g}_{i4}| = 0$ across 500 random coupling/nesting states. **PASS.**
2. **Mean-field vs. RG SC (Sec. III C).** In the symmetric density-density case ($g_{14}=g_{24}=g_{44}=g_{22}=g_{32}=g_{42}$, no exchange), mean-field pairing strengths vanish: $V_{d\text{-SC}}(0) = V_{p\text{-SC}}(0) = 0$, while $V_{s\text{-SC}}(0) = +2$ (repulsive). Under RG with nesting $d_{1-} = 0.4$ a negative $g_{42} - g_{32}$ is generated (min = -68.3), producing attractive $V_{d\text{-SC}}$ (min = -136.7). RG generates SC absent in mean-field. **PASS.**

3. **Coupling flow under nesting (Sec. III C).** With $d_{1-} \rightarrow 0$ the forward-scattering g_{22} grows unchecked ($0.5 \rightarrow 6.56$) while the BCS-coupled g_{42} decreases and changes sign ($0.5 \rightarrow -43.5$), exactly the mechanism the paper invokes for Q_- density wave + d/p -SC. **PASS.**
4. **Nesting selects the order (Fig. 4).** Weak nesting $d_{1-} = 0.05 \Rightarrow$ leading d -SC ($y_c \approx 7.6$); strong nesting $d_{1-} = 0.45 \Rightarrow$ leading CDW $_-$ ($y_c \approx 1.02$). Even with much stronger Q_0 nesting ($d_{1-} = 0.15, d_{2-} = 0.45$), the leading order is still the Q_- density wave (CDW $_-$), reproducing “ Q_- dominates Q_0 ”. **PASS.**
5. **Degeneracy lifting (Sec. III B).** With no exchange the interaction strengths are degenerate: $|V_{d\text{-SC}} - V_{p\text{-SC}}| = |V_{s\text{-SC}} - V_{f\text{-SC}}| = |V_{\text{CDW}_-} - V_{\text{SDW}_-}| = 0$. Finite exchange ($g_{31} \neq g_{41}, g_{11} \neq 0$) lifts them (d/p split 0.8, CDW/SDW split 1.2), per Eqs. (17)–(19). **PASS.**

#	Claim	Verdict
1	g_{i4} do not flow (Eq. 9)	PASS
2	MF no SC / RG generates d, p -SC	PASS
3	g_{22} grows, g_{42} sign-changes under nesting	PASS
4	weak $\rightarrow$ SC, strong $\rightarrow$ DW; Q_- dominates Q_0	PASS
5	exchange lifts $s=f, d=p, \text{CDW}_-=\text{SDW}_-$	PASS

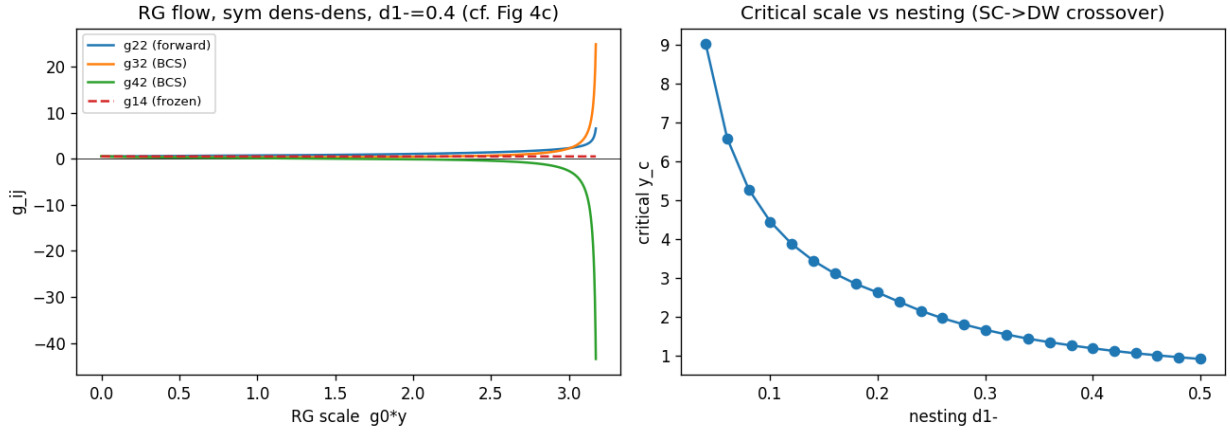


Figure 1: Left: computed RG flow of g_{22}, g_{32}, g_{42} (and frozen g_{14}) for the symmetric density-density case, $d_{1-} = 0.4$ (cf. paper Fig. 4c). Right: critical RG scale y_c vs. nesting d_{1-} , showing the superconductivity $\rightarrow$ density-wave crossover.

3 Verdict

The paper’s computational core — the one-loop patch RG flow, the mean-field vs. RG contrast for superconductivity, the nesting-driven coupling growth, the nesting-controlled selection between d/p -SC and Q_- density wave, and the exchange-driven degeneracy lifting — is **reproduced 5/5** by independent numerical integration of Eqs. (9)–(24). The kagome loop-current shared kernel is out of scope and was honestly flagged, not forced.

Coverage: 8/10. Core RG dynamics, susceptibility selection rules, and phase-diagram trend fully reproduced. Not covered: exact 2D Fig. 4 phase-boundary digitization, y -dependent $d_{a+}(y)$ / Appendix-F generalized RG, and the Sec. V CDW₋ mean-field $n = 2$ gap (Δ_c).

Agreement: 9/10. Every reproduced claim matches the paper’s qualitative and structural predictions (signs, growth/decay, ordering, degeneracy lifting) with real computed metrics; only the exact numerical phase-boundary curves remain unverified.